Julius Zhang

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Education

Courant Institute of Mathematical Sciences, New York University, computer science *PhD.* starting 2026.9

- Verifiable systems and theoretical computer science.

Stanford University, computer science *M.S.* and mathematics *B.A.* 2019.9 - 2024.6

- General Yaowu Wang Scholarship.

St Paul's School 2017.9 - 2019.6

- Full scholarship.

Talks

- **p-Ordinary Cohomology for $\mathrm{SL}(2)$ over Number Fields (after Hida)**,
  Hida Theory Seminar, Columbia University, 2026.

Selected Papers and Manuscripts

- **The Trace-Zero Subgroup and the Relative Trace for BN254**,
  Julius Zhang,
  Research Note, 2026.
  pdf
- **Complex Cobordism and Formal Group Law**,
  Julius Zhang,
  Manuscript (Undergraduate Honours Thesis), Stanford University, 2024.
  pdf

Experience

a16z Researcher, 2025.4 - Present

- Design and build Jolt, an open-source zero-knowledge virtual-machine in Rust.
- Proof compression research: designed torus compression system and proved theoretical optimality via Hilbert 90.
- Audited elliptic-curve optimizations, formalizing folklore cryptographic results into rigorous proofs.
- Performance engineering across the proving stack, including sum-check optimization.

Millennium Researcher, 2024.6 - 2025.4

- Convex optimization and robust statistical estimation for real-time trading systems.

Stellar Development Foundation Consultant, 2023.6 - 2023.9

- Byzantine fault tolerance, C++, stellar-core.
- Performance optimization: improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.

Distributed Systems and TCS Research, Stanford University Researcher, 2023.9 - 2024.6

- Under David Mazières, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules.
- Research PCP under Aviad Rubinstein, presented full proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.

Topology Research, Stanford University Researcher, 2020.7 - 2022.5

- Honours thesis related to Floer homotopy theory under Mohammed Abouzaid (see Papers). Proved a result sketched by Daniel Quillen, Fields medalist, relating the Gysin homomorphism of complex cobordism to the Grothendieck residue symbol. Produced a recursive formula to compute quantum product correction term in Gromov-Witten invariant valued cohomology.
- Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.

Skills

Computer Science:Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, Floer theory, gauge theory, concentration inequalities.